ULUSLARARASI KIBRIS ÜNİVERSİTESİ
CYPRUS INTERNATIONAL UNIVERSITY

Project Title: CAR RENTAL SYSTEM

CMPE314 – SOFTWARE ENGINEERING

PROJECT PROPOSAL

Submitted To: Okt.Behnam Shahrıarı

Group Number: 1

Submitted By: Alp Batuhan Bilgin

Alp Batuhan Bilgin - 22207903

Mert Yorulmaz - 22314700

Fatmatul Zehra Tas - 22208075

Date of Submission: 22/03/2026

1. INTRODUCTION

1.1. Project Background

Car rental companies often manage their vehicles, rental operations, and customer data using manual processes or outdated systems. This approach makes it difficult to efficiently track vehicle availability and leads to operational inefficiencies. Therefore, there is a need for a modern, centralized system to streamline these processes.

1.2. Problem Statement

There is no centralized system that efficiently manages vehicle availability, rental transactions, and return processes. This results in errors, delays, and difficulties in managing both vehicles and customer interactions.

1.3. Project Aim

The aim of this project is to develop a car rental management system that tracks vehicles, manages rental and return processes, and calculates late return fees efficiently.

2. PROPOSED SOLUTION

2.1. Description of the System

This project proposes a backend-oriented system designed to manage car rental operations. The system will track vehicle availability, manage rental transactions, handle return processes, and calculate late fees. All operations will be controlled through an administrative interface, ensuring efficient and centralized management.

2.2. Key Features (High-Level)

- User authentication (Admin/Staff login)
- Vehicle management (add, update, delete vehicles)
- Vehicle availability tracking
- Rental transaction management
- Return processing
- Late fee calculation
- Reporting system
- Customer feedback and rating system

3. STAKEHOLDERS AND TARGET USERS

3.1. Primary Users

- **Admin / Staff:** Responsible for managing vehicles and rental operations efficiently.
- **Customers (Indirect Users):** Benefit from faster, more reliable, and organized rental services.

3.2. Secondary Stakeholders

- **System Administrator:** Responsible for system maintenance, updates, and security.

4. REQUIREMENT HIGHLIGHTS

ID	Requirement Description	Priority (High/Med/Low)
FR-01	The system shall provide user authentication (Login) for Admin and Registered Customers..	High
FR-02	The system shall allow adding, updating, and removing vehicles from the inventory.	High
FR-03	The system shall display real-time vehicle availability status.	High
FR-04	he system shall support car searching and viewing detailed vehicle information.	High
FR-05	The system shall support rental transaction (Make Reservation) and cancellation.	High
FR-06	The system shall process payments through an external Payment Gateway.	HIgh
FR-07	The system shall send automated email confirmations after successful transactions.	Medium
FR-08	The system shall support vehicle return processing and calculate late fees.	High
FR-09	The system shall generate operational and inventory reports for Admin.	Low
FR-10	The system shall allow users to provide feedback and apply discount codes.	Low

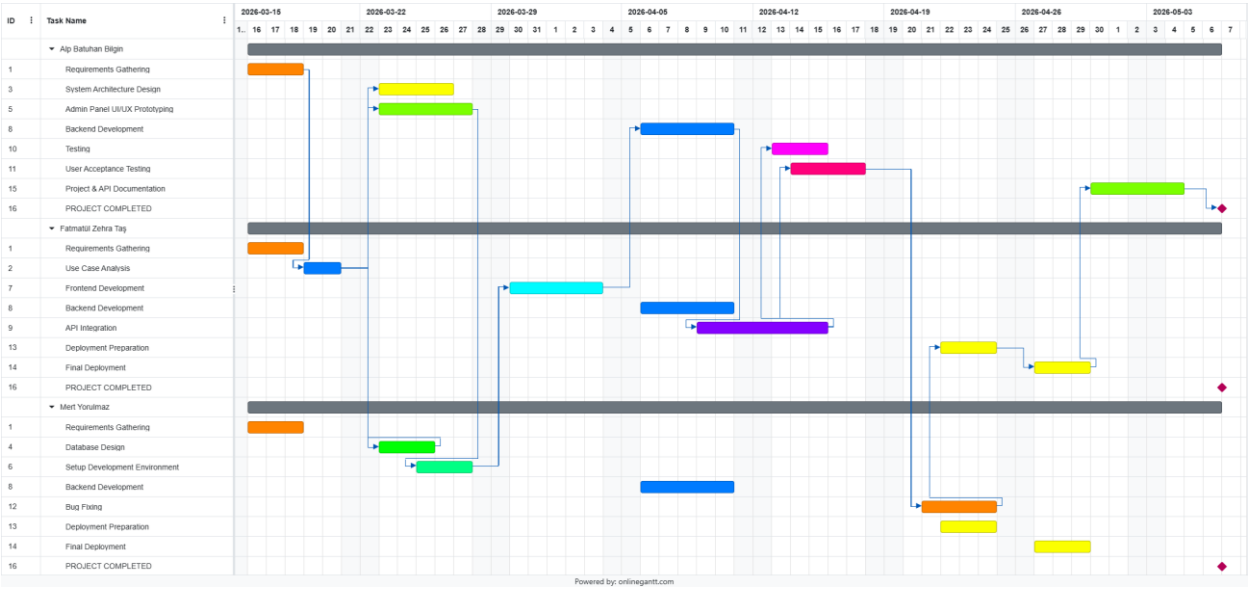
5. TECHNOLOGY STACK

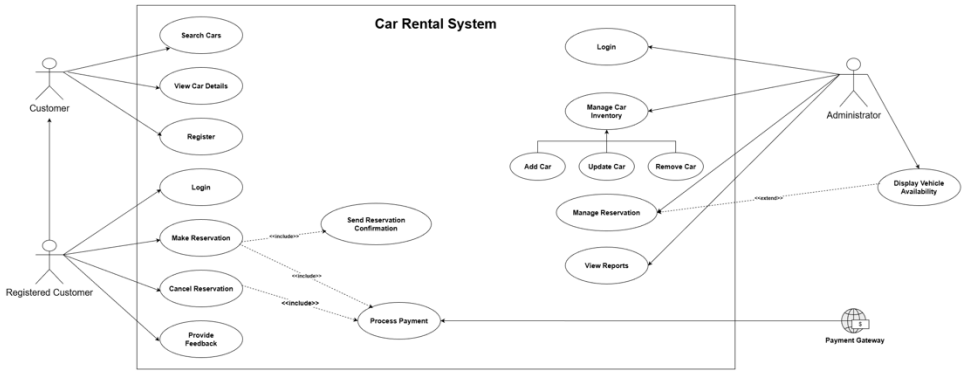
Layer	Technology Choice	Justification
Frontend (UI)	C# Windows Forms	Provides a user-friendly and efficient desktop interface for system management.
Backend (Logic)	C# (.NET Framework)	Enables robust business logic implementation with strong object-oriented programming support.
Database	Microsoft SQL Server LocalDB	Lightweight, easy to configure, and fully compatible with the .NET ecosystem.
Version Control	Git / GitHub	Supports team collaboration, version tracking, and efficient code management.
Open Source Libraries	.NET Built-in Libraries	Accelerates development using reliable and pre-integrated components..

6. PROJECT PLAN AND TIMELINE

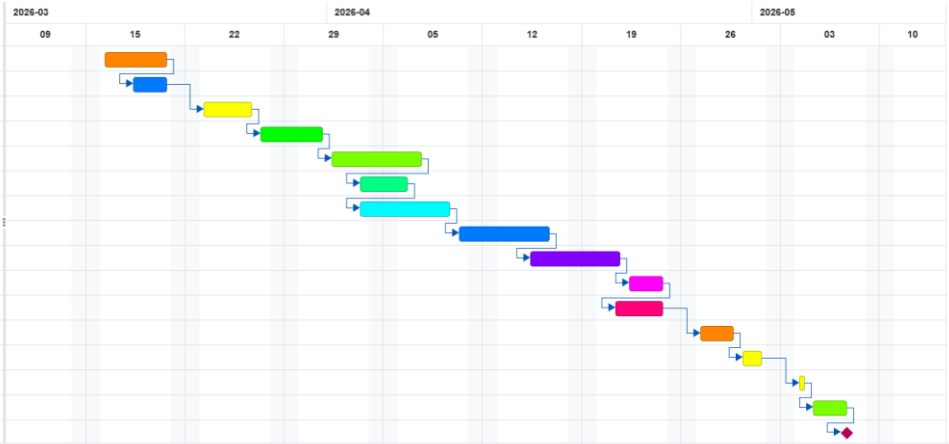
6.1. Task Breakdown & Gantt Chart

Phase	Week	Activities	Deliverable
Phase 1	Week 1	Requirements Analysis	SRS Draft
Phase 2	Week 2	Use Case Analysis	Use Case Diagrams
Phase 3	Week 3	System Design & Database Design	UML Diagrams & Database Schema
Phase 4	Week 4–5	Frontend & Backend Development	Functional System
Phase 5	Week 6	API Integration & Testing	Tested System
Phase 6	Week 7	Bug Fixing & Deployment	Final Version
Phase 7	Week 8	Documentation	Final Report





ID	Task Name
1	Requirements Gathering
2	Use Case Analysis
3	System Architecture Design
4	Database Design
5	Admin Panel UI/UX Prototyping
6	Setup Development Environment
7	Frontend Development
8	Backend Development
9	API Integration
10	Testing
11	User Acceptance Testing
12	Bug Fixing
13	Deployment Preparation
14	Final Deployment
15	Project & API Documentation
16	PROJECT COMPLETED



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6.2. Team Responsibilities

Team Member	Responsibilities
Alp Batuhan Bilgin	Project management, risk management, testing, user acceptance testing
Mert Yorulmaz	Backend development, database design, system architecture
Fatmatül Zehra Taş	UI/UX design, frontend development, documentation

7. CONCLUSION

This project aims to provide a structured and efficient solution for managing car rental operations. By digitizing the process, it reduces errors, improves efficiency, and enhances overall system usability. Furthermore, the project demonstrates the practical application of software engineering principles in solving real-world problems.

GITHUB LINK

https://github.com/22314700/CMPE314-Software-Engineering-Project